

Robust signal

$k_{\text{bio}}: 0.025 \rightarrow 0.027 \rightarrow 0.032 \text{ d}^{-1}$

Validated

model vs days 5-25, $R^2 \approx 0.95$

Batch→column

$\times 0.30\text{--}0.41$ (59–70% lower)

Bench-grounded

vG $\alpha=0.145$, $k_{\text{t}}a=0.11 \text{ h}^{-1}$